

Materials

1. Data sources and acquisition

Table 1: dataset, access date and other info

Dataset	Purpose	Access date	Filtering	Source
Uniprot	Human Protein sequences + protein annotation	Feb 2025	Reviewed sequences + proteins with valid Uniprot ids	https://www.uniprot.org/
gnomAD v2.1.1 leftover to GRCh 38	SNPs	2025	SNPs in exon regions; coding SNPs; AF >0.1; Filter = PASS; population = EUR*	https://gnomad.broadinstitute.org/
Ensembl v110	VEP indexed cache	2025	https://ftp.ensembl.org/pub/release-110/variation/indexed_vep_cache/homo_sapiens_vep_110_GRCh38.tar.gz	
HPA v24	HPA RNA dataset	2025	Valid cross-reference with Uniprot-ensembl IDs	https://www.proteinatlas.org/download/rna_tissue_hpa.tsv.zip
GTEX v8	Transcript-level expression	2025		https://storage.googleapis.com/adult-gtex/bulk-gtex/v8/rna-seq/GTEX_Analysis_2017-06-05_v8_RSEMv1.3.0_transcript_tpm.gz
IEDB	Peptides annotations	2025	Human; linear peptide; peptide sequence	https://www.iedb.org/
	term = ... (list of peptide sequences) base_uri='https://query-api.iedb.org' search_params = {'linear_sequence': term, 'structure_type': 'eq.Linear peptide', 'host_organism_iri': 'eq.NCBITaxon:9606', 'source_organism_iri': 'eq.NCBITaxon:9606', 'order': 'linear_sequence', 'offset': 0}			
PEPREP	Peptides annotations	2025	Human; peptide sequence	BioCopy GmbH

Table 2: tools and packages that were used directly in EpiTox

Tools/package	Purpose	Reference
BLASTp	Sequence search	Mahram et al., <i>ACM (TSETS)</i> , 2015
PEPMatch	Sequence search	Marrama et al., <i>BMC bioinformatics</i> , 2023
tabix 1.9	SNPs extraction*	Danecek et al., <i>Bioinformatics</i> , 2021
Bcftools		
vep	SNPs annotation	Hunt et al., <i>Human mutation</i> , 2022
NetMHCpan	HLA binding predictions	Reynisson et al., <i>Nucleic Acids Research</i> , 2020
MHCflurry		O'Donnell et al., <i>Cell Systems</i> , 2020
R packages		
ggplot2	Visualization	H. Wickham. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2016.
lessR		Gerbing D (2025). lessR: Less Code, More Results. R package version 4.4.5, < https://CRAN.R-project.org/package=lessR >.
		Gerbing DW (2021). “Enhancement of the Command-Line Environment for use in the Introductory Statistics Course and Beyond.”
		Journal of Statistics and Data Science Education, 29(3), 251-256. doi:10.1080/26939169.2021.1999871
		< https://doi.org/10.1080/26939169.2021.1999871 >.
biomaRt	Cross-reference and gene annotation	Mapping identifiers for the integration of genomic datasets with the R/Bioconductor package biomaRt. Steffen Durinck, Paul T. Spellman, Ewan Birney and Wolfgang Huber, Nature Protocols 4, 1184-1191 (2009). BioMart and Bioconductor: a powerful link between biological databases and microarray data analysis. Steffen Durinck, Yves Moreau, Arek Kasprzyk, Sean Davis, Bart De Moor, Alvis Brazma and Wolfgang Huber, Bioinformatics 21, 3439-3440 (2005).
rmarkdown	Report generation	Allaire J, Xie Y, Dervieux C, McPherson J, Luraschi J, Ushey K, Atkins A, Wickham H, Cheng J, Chang W, Iannone R (2024). rmarkdown: Dynamic Documents for R. R package version 2.29, < https://github.com/rstudio/rmarkdown >.
tidyr	<i>Data manipulation and analysis</i>	Wickham H, Vaughan D, Girlich M (2024). tidyr: Tidy Messy Data. R package version 1.3.1, < https://CRAN.R-project.org/package=tidyr >.
Python modules		

Panda	<i>Data manipulation and analysis</i>	Wes McKinney, Proceedings of the Python in Science Conference, 2010
scikit-learn (sklearn)	<i>Machine learning algorithms and utilities</i>	Pedregosa et al., Journal of Machine Learning Research, 2011
NumPy	<i>Numerical computing, arrays, linear algebra</i>	Harris et al., Nature, 2020
Bio.SeqUtils (Biopython)	<i>Sequence utility functions (molecular weight, composition, etc.)</i>	Cock et al., Bioinformatics, 2009
SciPy	<i>Scientific computing, optimization, statistics</i>	Virtanen et al., Nature Methods, 2020
peptides	<i>Peptide analysis (hydrophobicity, pI, composition, etc.)</i>	Wirth & Nystrom, Journal of Open Source Software, 2020
Framework		
R 4.4.1	R Core Team (2024). <code>_R: A Language and Environment for Statistical Computing_</code> . R Foundation for Statistical Computing, Vienna, Austria. < https://www.R-project.org/ >.	
Python 3.9.16	Van Rossum, G., & Drake, F. L. (2009). Python 3 Reference Manual. Scotts Valley, CA: CreateSpace.	
Conda		
snakemake 7.24.2	Mölder, F., Jablonski, K.P., Letcher, B., Hall, M.B., Tomkins-Tinch, C.H., Sochat, V., Forster, J., Lee, S., Twardziok, S.O., Kanitz, A., Wilm, A., Holtgrewe, M., Rahmann, S., Nahnsen, S., Köster, J., 2021. Sustainable data analysis with Snakemake. F1000Res 10, 33.	
*we followed gnomAD population’s description for selection (see https://gnomad.broadinstitute.org/news/2017-02-the-genome-aggregation-database/)		

2. PEPREP: in-house database for pHLA annotations

Along with public resources, EpiTox uses in-house established resource for experimentally validated pHLA complexes, peptides repertoire (PEPREP). PEPREP aggregates and harmonizes public and propriety data to provide different information for each pHLA, for example HLA-allele, disease, and tissue.

3. Known peptides/off-targets in MAGE-A3 context

Three studies and one patent were used to create a list of 18 peptides that was used as disclosed or known off-targets for MAGE-A3 case. The peptides sequences and annotations were incorporated as they were provided by the papers.

Table 3: known peptides in MAGE-A3 context as speculated or confirmed off-targets

Gene name	peptide	mismatch	source
MAGEA9, MAGEA3	KVAELVHFL	0	PMID23377668, PMID21149604, WO2021173674A1

DDX28	KVAELVHIL	1	PMID23377668, WO2021173674A1, PMID27439771
MAGEA12	KMAELVHFL	1	PMID23377668, PMID21149604, WO2021173674A1, PMID27439771
MAGEA6	KVAKLVHFL	1	PMID21149604, WO2021173674A1
MAGEA8	KVAELVRFL	1	PMID21149604, WO2021173674A1
MAGEC2	KVAELVEFL	1	PMID21149604
MAGEC3	KVAELVQFL	1	PMID23377668, WO2021173674A1, PMID27439771
EPS8L2	SAAELVHFL	2	WO2021173674A1, PMID27439771
MAGEA1	KVADLVGFL	2	PMID21149604, WO2021173674A1
MAGEA2	KMVELVHFL	2	PMID21149604, WO2021173674A1
MAGEA4	KVDELAHFL	2	PMID21149604, WO2021173674A1
MAGEA5	KVADLIHFL	2	WO2021173674A1
MAGEB18	KVVSLVHFL	2	PMID23377668, WO2021173674A1
MRV11, MRVI1	KLEELVHFL	2	PMID23377668, WO2021173674A1, PMID27439771
EPS8	SAADLVHFL	3	PMID23377668, WO2021173674A1, PMID27439771
MAGEA10	KVTDLVQFL	3	WO2021173674A1
MAGEF1	TVaelVQFL	3	PMID23377668, WO2021173674A1
PPP2R1B	GIAELVHFS	3	PMID23377668, WO2021173674A1
MAGEA11	KIIDLVHLL	4	WO2021173674A1

4. EpiTox architecture

4.1 PrediTopes: HLA binding prediction and experimental evidence

4.1.1 Data structure

While quantitative binding measurements exist for some peptides (e.g., IC50 binding affinities, raw MS values), categorical evidence (validation: yes/no; binding: strong/intermediate/weak; class: I/II) provides more complete coverage across the dataset. We therefore utilized categorical evidence as the primary input, producing discrete posterior probabilities when evidence combinations recur. The framework accommodates future integration of quantitative data where available.

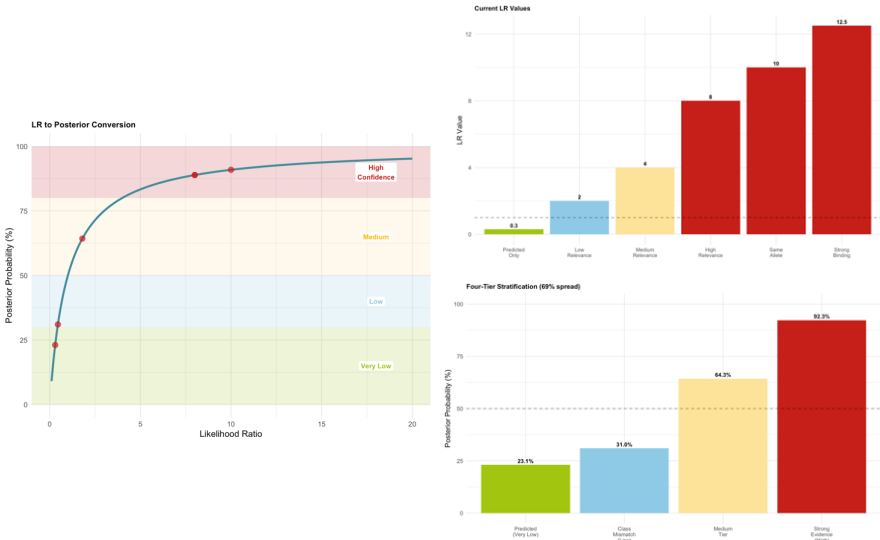
Furthermore, experimental data were aggregated for each protein-peptide pair (uniprot id – peptide sequence), preserving allele-binding relationships and associated metadata including tissue/disease context, binding affinity measurements, and study quality indicators. This protein-specific aggregation ensures that evidence assessments reflect the exact biological context of each peptide-source.

Table 4: Likelihood ratios used in Bayesian framework

Evidence quality:	
Multiple high-relevance studies: target allele + tissue context,	LR = 12
Single high-relevance study	LR = 8
Multiple/single medium-relevance studies	LR = 6/4
Multiple/single low-relevance studies	LR = 3/2
No experimental evidence	LR = 0.3 (conservative penalty)
HLA allele specificity:	
Target allele match	LR = 10
Different allele	LR = 2
HLA class match (when allele is unknown):	Same: LR = 2 Different: LR =0.5
Binding affinity:	
Base LR: strong = 5, intermediate = 3, weak = 2, non-binder = 1	
Target-specific multipliers: strong = 2.5x, intermediate = 1.5x, weak = 1x, negative = 0.2x	
Tissue/disease context:	
Normal tissue	LR = 10
Same disease (optional)	LR = 6
Similar tissue	LR = 3

Supplementary figure 1

LR sensitivity analysis



Results and methods

Table 5: Position-specific HLA-binding of MAGEA3

Position	AA	Class	FC_mean	Sensitivity	% Better than target
P1	K	Classical	5.96	39	15.79
P2	V	Classical	24.41	108.67	15.79
P3	A	Permissive	1.63	3.9	31.58
P4	E	Neutral	1.91	2.8	5.26
P5	L	Permissive	1.12	0.6	31.58
P6	V	Neutral	1.43	1.46	15.79
P7	H	Neutral	1.62	2.65	0
P8	F	Neutral	1.5	1.8	5.27
P9	L	Classical	305.99	1135.26	10.53

AA: Target sequence amino acid wise.
FC_mean: mean fold change in KD upon substitution.
Sensitivity: normalized range of KD change.
% Better than WT: percentage of substitutions with better KD than target.

Supplementary figure 2

Target positional template (TPT)

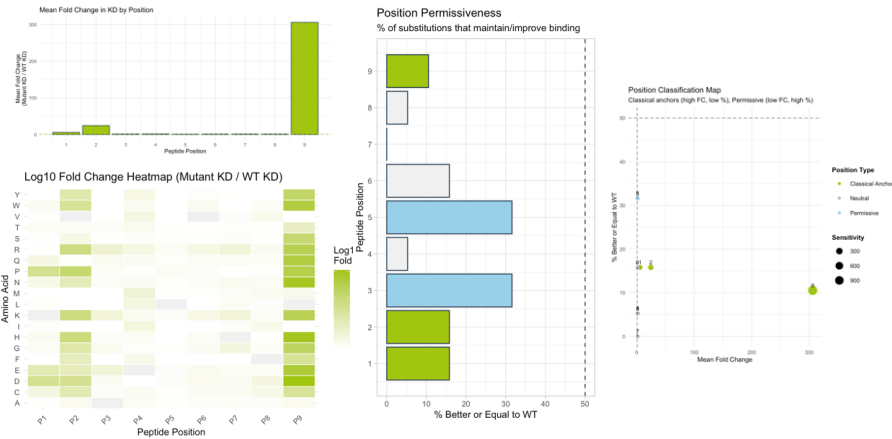


Table 6: Identified off-target candidates from different MAGE families (A, B, C, F & E)

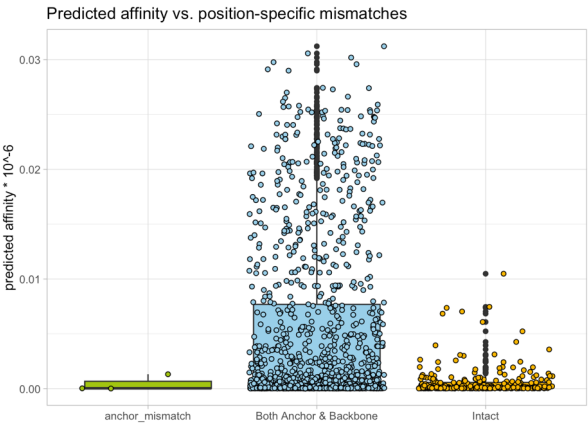
uniprot	Gene Name	Peptide	Blosum62 similarity	Edit distance	Wildtype	HLA-A*2:01 Affinity
MAGEA family						
P43357	MAGEA3	KVAELVHFL	44	0	Yes	14,3961329
P43362	MAGEA9	KVAELVHFL	44	0	Yes	14,3961329
P43365	MAGEA12	KMAELVHFL	41	1	Yes	10,8250373
P43361	MAGEA8	KVAELVRFL	36	1	Yes	52,6518232
P43360	MAGEA6	KVAKLVHFL	40	1	Yes	23,7568242
P43356	MAGEA2	KMVSLVHFL	37	2	Yes	10,885883
P43355	MAGEA1	KVADLVGFL	31	2	Yes	36,2984481
P43358	MAGEA4	KVDELAHFL	34	2	Yes	29,9630256
P43363	MAGEA10	KVTDLVQFL	29	3	Yes	30,3629588
P43364	MAGEA11	KIIDLVHLL	29	4	Yes	15,3669415
Q7L5Y9	MAEA (PIG5)	VVAELEKTL	14	4	Yes	1046,11964
Q7L5Y9	MAEA (PIG5)	VVAELQKTL	14	4	No	967,514188
MAGEB family						
Q96M61	MAGEB18	KVVSLVHFL	35	2	Yes	19,2521481
A8MXT2	MAGEB17	KTGELVQFL	28	3	Yes	104,838719
A2A368	MAGEB16	KVAFLVNFM	27	3	Yes	1158,76239
Q96LZ2	MAGEB10	KVIILVHYL	28	3	Yes	38,8062522
O15479	MAGEB2	KSGSLVQFL	21	4	Yes	1673,43194
O15481	MAGEB4	KTKMLVQFL	20	4	Yes	7170,88466
O15480	MAGEB3	KTNMLVQFL	19	4	Yes	1665,05912
Q9BZ81	MAGEB5	KVGILLEQFL	18	4	Yes	2413,8697
A8MXT2	MAGEB17	KVLEFVAKL	16	4	Yes	20,0630444
MAGEC family						
Q9UBF1	MAGEC2	KVAELVEFL	36	1	Yes	19,868702
Q8TD91	MAGEC3	KVAELVQFL	36	1	Yes	20,4910337
O60732	MAGEC1	KVDELARFL	26	3	Yes	111,874657
Q8TD91	MAGEC3	KVDKLVQFL	26	3	Yes	69,3400635
MAGEF family						
Q9HAY2	MAGEF1	TVAELVQFL	30	2	Yes	40,0224415
MAGEE family						
Q9HCI5	MAGEE1	NVAELLQFL	28	3	Yes	57,7469841

Kommentiert [HAH1]: is this needed in the main text or could be shifted to supplementary?

Kommentiert [SS1R2]: supplementary

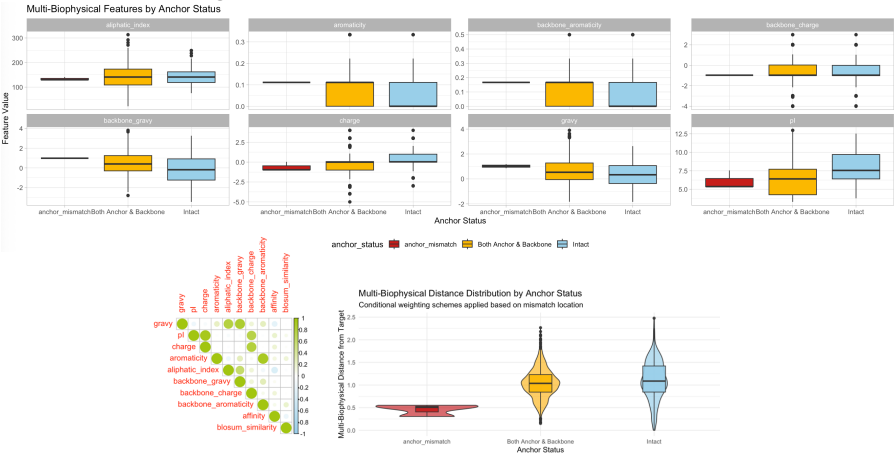
Supplementary figure 3

Position-specific mismatches vs. predicted binding affinity



Supplementary figure 4

Multi-features ranking score



Supplementary figure 5

Bi-features vs. multi-features assessment

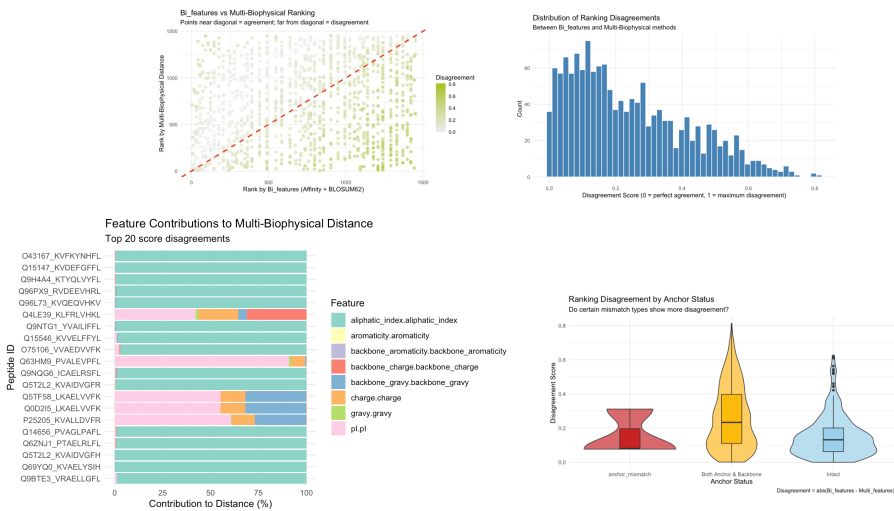


Table 5: SNP-derived peptide ranked by different scores

Multi-features	Bi-features		
	High	Moderate	Low
2-4			
High	9	15	2
Moderate	1	4	1
Low	0	3	0